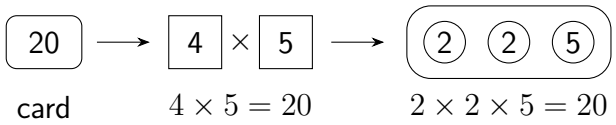
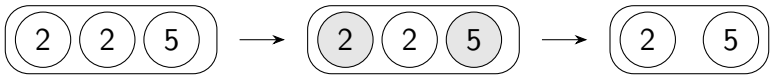
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 with no factors except 1 and itself. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Equal tokens are interchangeable.

Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



value 20 choose shaded tokens value 10

$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 5: Draw prime bags using any of 2, 3 and 5 whose values have exactly 6, 8, 9, 10 and 12 divisors. Find all possible token-count patterns for each target. Bags with two 2s and one 3, or two 5s and one 2, have the same pattern.

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors

Problem 6: Find the least positive number with each divisor count below. You may use any primes. Explain how you can rule out every smaller number.

4 divisors

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors